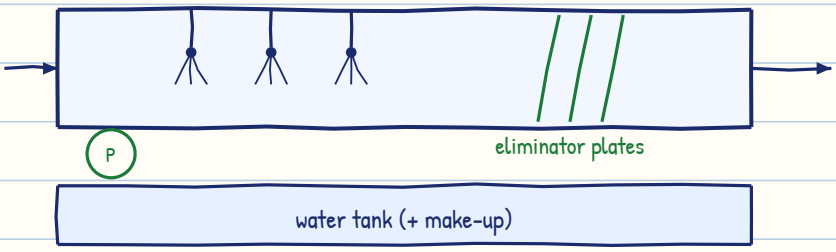
Syllabus: Air conditioning equipment – humidifiers, dehumidifiers, air filters, fans and blowers. Human comfort – requirements of temperature and humidity, effective temperature, comfort chart. Heat pump – heat sources, different heat pump circuits. Energy efficient AC systems – star rating.

I. Humidifiers

- A **humidifier** adds moisture to air (increases W) – needed in winter air-conditioning, textile mills, printing, paper stores, hospitals, and museums.

(a) Air-washer / spray humidifier

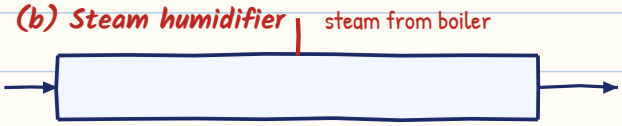


Spray: water droplets evaporate in air → W ↑, DBT ↓ (adiabatic)

Steam: W ↑, DBT ≈ const (isothermal), hygienic

$m_s = m_a (W_2 - W_1)$

(b) Steam humidifier



Types

- **Spray / air washer:** water sprayed by nozzles into the air stream; evaporative (adiabatic) – air cools while W rises. Recirculating pump, tank, eliminators to catch droplets. Also cleans the air.
- **Steam humidifier:** steam from a boiler or electrode / resistance boiler injected into the duct. Almost **isothermal**, hygienic, accurate control; costly energy.
- **Evaporative pad / pan humidifier:** wetted pad, air blown across; used in small units.
- **Ultrasonic humidifier:** piezo-crystal vibrates at 1.7 MHz, breaking water into micro-droplets (mist). Low power; needs demineralised water.
- **Capillary / wetted-media, centrifugal atomising** – domestic types.

Steam needed $m_s = m_a (W_2 - W_1)$ kg/s ; heat added ≈ m_s × 2676 kJ/kg (for steam at 100 °C)

Example

2 kg/s of air is humidified from W = 0.006 to 0.009: m_s = 2 × 0.003 = **0.006 kg/s = 21.6 kg/h** and heat = 0.006 × 2676 = **16 kW**.

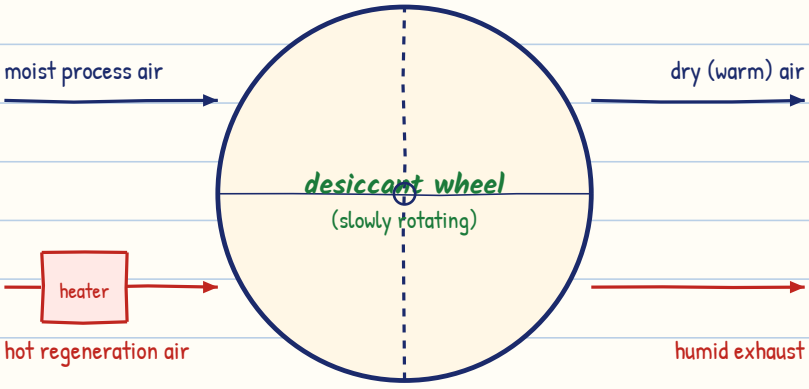
Note: Humidity control by humidistat; keep RH > 30 % in winter (static electricity, dry skin) and < 60 % (mould).

2. Dehumidifiers

- Removing moisture (reducing W) is needed in humid climates, cold storage, pharma, electronics, ships, basements and where closely controlled RH is required.

Methods

- **Cooling and dehumidification:** pass air over cooling coil below its DPT – water condenses (used in all ACs and **refrigerant dehumidifiers**). Air is then **reheated** if needed. Low energy at moderate humidity, poor below ≈ 40 % RH.
- **Chemical (sorption) dehumidification:** hygroscopic material absorbs / adsorbs the vapour.
  - **Absorbent (liquid):** lithium chloride, triethylene glycol sprays – the vapour is absorbed; liquid regenerated by heating.
  - **Adsorbent (solid):** silica gel, activated alumina, molecular sieve, zeolite – vapour held on surface; regenerated by hot air (100-130 °C).
- **Compression:** compressing the air increases p_v → water condenses (compressed air driers).



Adsorption on silica gel, molecular sieve or LiCl.
Process side: W ↓, DBT ↑
Regeneration: 60 – 130 °C
hot air drives off moisture

Method	Process line	Remarks
Cooling coil	W ↓ and DBT ↓	Common, needs reheat
Chemical (adsorption)	W ↓, DBT ↑ (h ≈ const)	Very dry air, 10 % RH possible
Desiccant wheel + cooling	Hybrid	Saves energy, uses waste heat

3. Air Filters

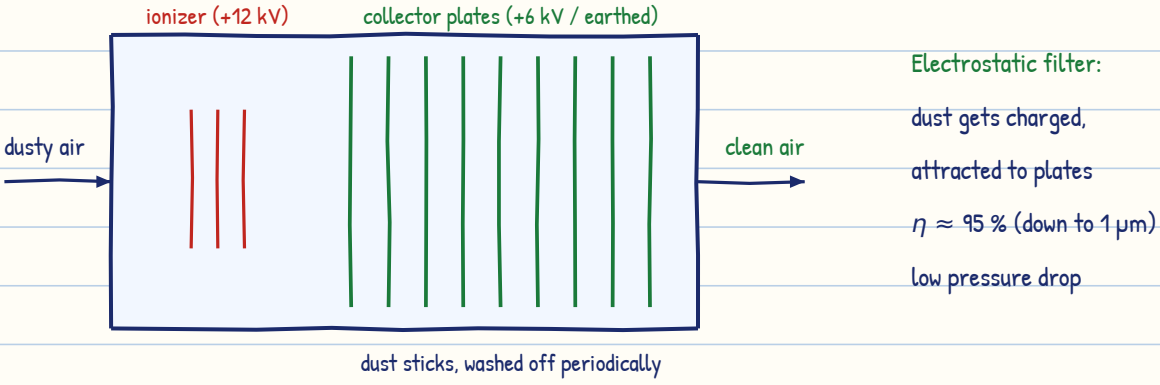
- Purpose: remove **dust, pollen, smoke, bacteria, fungus, odour** from supply / return air – to protect health, occupants, equipment and process.

Terms

- **Efficiency** – percentage of particles of a given size removed. **Arrestance** – weight of arrested dust. **Dust-holding capacity, pressure drop** (initial 50-100 Pa → final 250 Pa; change filter), **MERV rating (1-16)**.

Types of air filters

- **Dry (fibre / panel) filters:** glass fibre, cotton, synthetic mat, pleated paper; throw-away or washable; cheap, pre-filter (efficiency 20-80 % at > 5 m). Face velocity 1-2.5 m/s.
- **Viscous-impingement (oil-coated) filters:** metal mesh or foam coated with oil; dust sticks to the oil film; washed and re-oiled; used in industrial / kitchen.
- **Wet filters (air washers):** water spray / scrubber – removes dust and soluble gases; also humidifies.
- **Electrostatic precipitator (ESP):** dust particles charged by corona, collected on oppositely charged plates; very high efficiency, low pressure drop, high initial cost and ozone risk.



- **HEPA (High Efficiency Particulate Air) filter:** glass micro-fibre pleated paper, **99.97 % at 0.3 m**. Cleanrooms, hospital operating theatres, pharma, labs. ULPA is 99.999 % at 0.12 m.

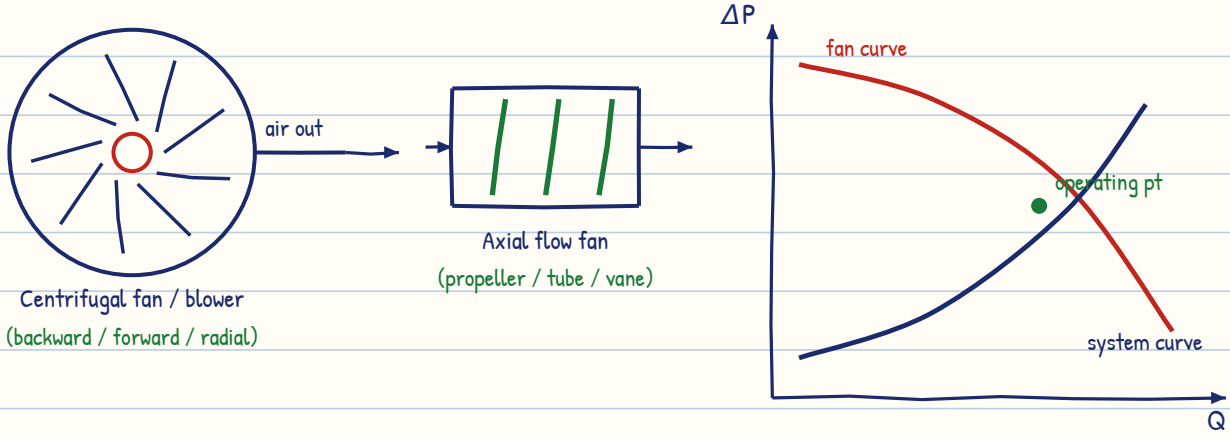
- **Activated carbon filters:** adsorb odours, gases, VOCs (kitchen, smoke).

Class	Typical efficiency	Use
Pre-filter (G3-G4)	60-90 % (> 10 m)	All AHUs, first stage
Fine filter (F7-F9)	80-95 % (1 m)	Offices, hospitals
HEPA (H13/H14)	99.95-99.995 % (0.3 m)	Cleanrooms, O.T.
Electrostatic	90-98 %	Industrial, hospitals

Note: Filter staging: pre-filter → fine filter → HEPA (final). Never skip the pre-filter – it protects the costly filters.

4. Fans and Blowers

- Fans supply air through ducts with small pressure rise (up to ≈ 1.1 pressure ratio, ~ 1-3 kPa); **blowers** give higher pressure (up to ≈ 10-15 kPa); compressors beyond.



Classification

- **Centrifugal (radial-flow):** air enters axially and leaves radially. Blades: **forward-curved** (small, quiet, low speed, used in FCU, AHU, power ↑ with flow – overloading), **backward-curved / airfoil** (highest efficiency, non-overloading, large AHU), **radial** (rugged, dusty air).
- **Axial flow:** air flows parallel to the shaft. **Propeller** (low pressure, large flow – ventilation), **tube-axial** (with duct casing), **vane-axial** (with guide vanes, high efficiency).
- **Cross-flow (tangential):** long cylindrical impeller – indoor split AC units, quiet.

Fan type	Flow	Pressure	Efficiency	Use
Propeller	Very high	Very low	40-50 %	Exhaust, cooling tower
Tube / vane-axial	High	Medium	70-85 %	Duct ventilation
Forward-curved	Medium	Medium	60-70 %	FCU, packaged AC
Backward-curved	High	High	80-90 %	Large AHU

Fan performance

Air power = $Q\,\Delta P$; Shaft power $P = Q\,\Delta P / \eta_{fan}$ (Q in m^3/s, ΔP in Pa → W)

Fan laws (same fan, speed N): $Q \propto N$, $\Delta P \propto N^2$, Power $\propto N^3$

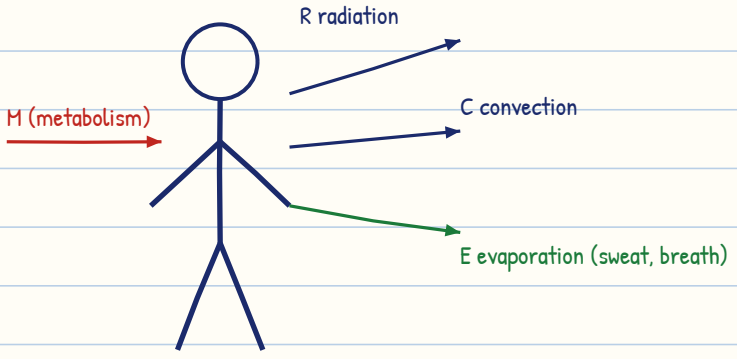
- Operating point is the intersection of **fan curve** and **system resistance curve** ($\Delta P \propto Q^2$). Control by dampers, inlet guide vanes or VFD (variable frequency drive – most energy efficient).

Example

- $Q = 2\ m^3/s$, $\Delta P = 500$ Pa, $\eta = 0.65$ → $P = 2\times500/0.65 =$ **1538 W ≈ 1.54 kW**.
- A fan gives $5\ m^3/s$, 400 Pa at 1000 rpm, 3 kW. At 1200 rpm: $Q = 6\ m^3/s$, $\Delta P = 400\times1.44 = 576$ Pa, power = $3\times1.728 =$ **5.18 kW**.

5. Human Comfort

- **Thermal comfort** is the condition of mind that expresses satisfaction with the thermal environment (ASHRAE 55). The body produces heat by metabolism (≈ 100 W for a seated person, 1 met = 58 W/m^2) and must lose it at the same rate to keep deep body temperature ≈ 37 °C.



Heat balance of body:

M - W = E + R + C + S

S = 0 → thermal comfort

S > 0 : feels hot

S < 0 : feels cold

- **Heat losses:** convection (air movement), radiation (to cooler surfaces), evaporation (sweat and respiration), conduction (negligible). Higher DBT and humidity → less loss → feels hot.

Factors affecting comfort

Factor	Effect
Air temperature (DBT)	Main; 22-26 °C comfortable
Humidity (RH)	High RH reduces sweat evaporation; optimum 40-60 %
Air velocity	0.1-0.25 m/s gentle; > 0.5 m/s draught; helps in summer
Mean radiant temperature	Radiation from walls, sun
Clothing (clo) and activity (met)	1 clo = 0.155 m^2K/W
Air quality, noise, light	Fresh air and cleanliness

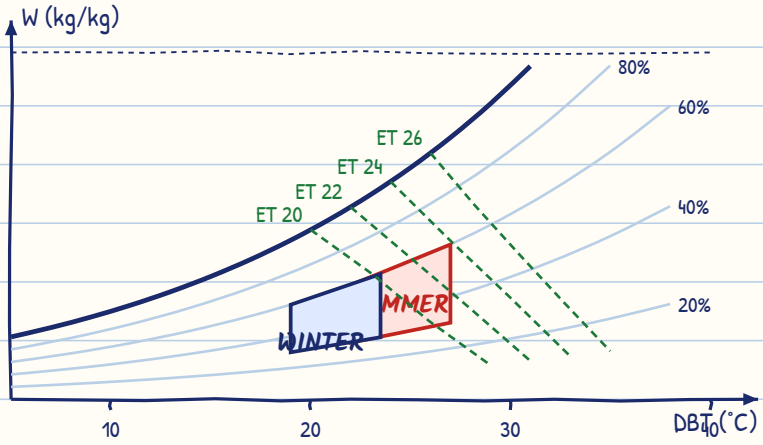
Requirements of temperature and humidity

- **Summer:** 24-26 °C DBT, 50-60 % RH (a little warmer to avoid too large indoor / outdoor difference – about 7-8 K). **Winter:** 20-22 °C, 40-50 % RH.
- Low RH (< 30 %) → dry mucous membranes, static; high RH (> 70 %) → clammy feeling, mould. Air velocity 0.15-0.25 m/s, vertical temperature difference < 3 K, fresh air.

Effective Temperature (ET)

- **Definition:** an index that combines the effects of DBT, humidity and air motion – ET is the temperature of **still, saturated air (RH = 100 %, V ≈ 0)** which produces the same sensation of warmth or cold as the given air state.
- Example: air at 26 °C and 50 % RH in still air has ET ≈ 22.5 °C → feels like saturated air at 22.5 °C.
- **Drawbacks:** over-estimates the effect of humidity at low temperatures; does not include radiation (corrected ET with globe temperature) and clothing/activity.

Comfort chart



Comfort chart (ASHVE)

ET lines slope down-right

(sat. air, still = ET)

Summer zone: ET ≈ 22–26 °C

Winter zone: ET ≈ 19–23 °C

RH 30 – 70 %

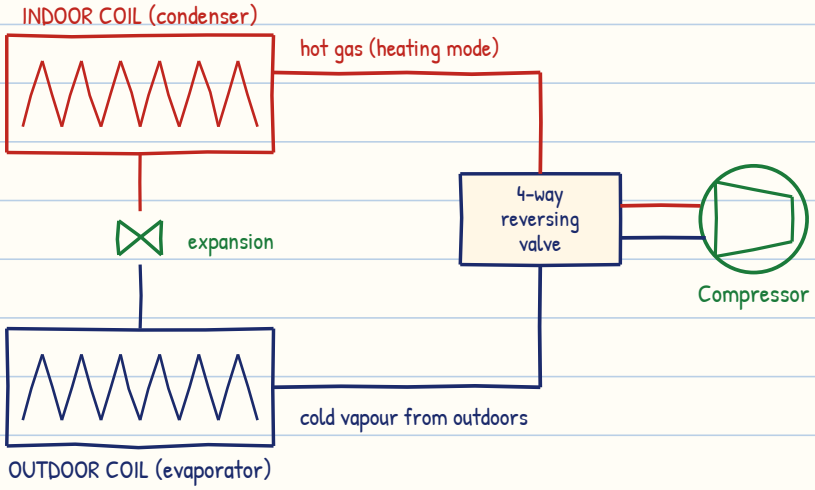
- The **ASHVE comfort chart** (Houghton and Yaglou) is the psychrometric chart on which lines of constant ET are plotted; the **comfort zones** for summer and winter, where ≥ 50 % of the persons are comfortable, are drawn.
- Using the chart: given DBT and RH → read ET; or choose the ET (say 23 °C) and find different combinations of DBT/RH giving the same comfort – helpful for choosing the AC design point.
- **Other indices:** Predicted Mean Vote (**PMV**, -3 to +3) and Predicted Percentage Dissatisfied (**PPD** ≥ 5 %) by Fanger; operative temperature; heat stress index (HSI); wet-bulb globe temperature (WBGT).

6. Heat Pump

- A heat pump is a refrigerating machine used for **heating**: it extracts heat Q_L from a low-temperature source (outside air, ground, water) and delivers $Q_H = Q_L + W$ to the space.

$$COP_{HP} = Q_H / W = COP_R + 1 \geq 1 \; ; \; \text{Carnot } COP_{HP} = T_H / (T_H - T_L)$$

- Typical COP 3–5 (more than 1 kW electric heating) – hence energy saving and lower CO_2.



Heating: indoor coil = condenser

Cooling: valve reverses, indoor coil = evaporator

$COP_{HP} = Q_H/W = COP_R + 1$

Defrost cycle for outdoor coil

- **Reversing valve** (4-way) interchanges the roles of the coils: in cooling mode indoor coil is the evaporator, outdoor coil the condenser. **Defrost** cycle is needed for outdoor coils at < 5 °C.

Heat sources (low-grade heat)

Source	Temperature	Merits	Demerits
Atmospheric air	−10 to 15 °C	Free, available anywhere	COP drops in cold; frost
Well / ground water	8–15 °C steady	High COP, steady	Needs wells, legal permission
Ground (soil) – horizontal / vertical loops	5–12 °C	Steady, best COP	High installation cost
Surface water (lake, river)	2–15 °C	Easy	Freezing risk
Solar (solar-assisted)	variable	Free energy	Needs storage
Waste heat (industry, exhaust air)	20–40 °C	Very high COP	Site specific

Heat pump circuits (classification by source and distribution)

Circuit	Source	Distribution	Example
Air-to-air	Outdoor air	Room air (ducts)	Reversible split AC, VRF
Air-to-water	Outdoor air	Hot water (radiators)	Domestic hot water, hydronic
Water-to-air	Water / ground loop	Room air	Ground source unit
Water-to-water	Water / ground	Hot water	Large buildings
Earth-to-air / earth-to-water	Soil	Air / water	Geothermal heat pump

- **Other cycles:** absorption heat pump (heat driven; gas fired), **heat pump water heater**, **heat recovery chiller** (simultaneous cooling and heating), **cascade** systems.

Example

Heating load 12 kW, COP = 4: W = 12/4 = **3 kW**, heat from source = 12 - 3 = **9 kW**. Carnot COP_{HP} between 0 °C and 22 °C = 295/22 = 13.4; actual COP 4 = 30 % of Carnot.

7. Energy Efficient AC Systems and Star Rating

- AC use is a huge part of peak electricity demand in India, so **BEE (Bureau of Energy Efficiency)** runs the **Standards & Labelling** programme with star labels (1 star – 5 stars: more stars = more efficient).
- Rating metric: **ISEER** (Indian Seasonal Energy Efficiency Ratio) = total seasonal cooling delivered (Wh) / total electrical energy consumed (Wh) over a typical Indian cooling season at weighted outdoor temperature bins (24-43 °C). It includes part-load operation, so **inverter ACs get a higher rating**.

> ISEER = Seasonal cooling output / seasonal electricity input (energy consumed ∝ 1/ISEER)

BEE star bands (split AC, valid 1 Jan 2026 – 31 Dec 2028)

Star	ISEER (split AC)	Window AC ISEER
1 star	3.50 - 3.79	2.90 - 3.09
2 star	3.80 - 4.29	3.10 - 3.29
3 star	4.30 - 4.99	3.30 - 3.49
4 star	5.00 - 5.59	3.50 - 3.69
5 star	≥ 5.60	≥ 3.70

Note: BEE revises these bands every few years (the 2026 update was tougher: an old 5-star is now about a 4-star). Always check the label year and ISEER value; verify the latest on the BEE website.

Features of energy efficient AC

- **Inverter (variable-speed) compressor** – matches capacity to the load; saves 30-50 % energy vs fixed-speed. **BLDC fans**, large copper heat-exchanger area, **R32 / R290** low-GWP refrigerant, electronic expansion valve, smart sensors, Wi-Fi, auto clean, dry mode.
- **Operational tips:** set the temperature at 24-26 °C (each 1 °C higher saves ≈ 6 % energy), regular filter cleaning, shading and insulation, correct tonnage, fans to circulate air, use timers.
- **System-level:** VRF, variable air volume, chilled-water plants with VFD pumps / fans, cooling-tower optimisation, thermal storage, heat recovery, building management system (BMS), evaporative pre-cooling of condenser air.

Example

Replacing a 3.6 ISEER AC by a 5.6 ISEER AC for the same cooling duty: energy saving = 1 - 3.6/5.6 = **35.7 %**. If the old unit consumed 1500 kWh per year, the new one uses about 964 kWh – saving 536 kWh.

7A. Numerical Problems and Short Answers

Problem 1 (humidifier with spray)

Air at 25 °C, W = 0.008 passes an air washer with 80 % efficiency and leaves almost saturated. Saturated W at the WBT ≈ 0.0125. Then W_2 = 0.008 + 0.8 (0.0125 - 0.008) = **0.0116 kg/kg**. For 3 kg/s of air, water evaporated = 3 × 0.0036 = **0.0108 kg/s = 38.9 kg/h**.

Problem 2 (dehumidification by cooling)

Air 1.2 kg/s at W_1 = 0.016 is cooled to W_2 = 0.009. Condensate = 1.2 × 0.007 = **0.0084 kg/s** ≈ **30 kg/h**. Latent load = 0.0084 × 2450 = **20.6 kW**.

Problem 3 (filter efficiency)

Upstream dust 20 mg/m^3, downstream 1 mg/m^3: efficiency = (20 - 1)/20 = **95 %**.

Problem 4 (body heat balance)

Seated person produces M = 104 W, does no external work (W = 0). Evaporation (E) = 25 W, convection (C) = 35 W, radiation (R) = 40 W → S = 104 - 100 = **+4 W** (slightly warm, body stores heat).

Problem 5 (fan power with VFD)

A fan of 3 kW at full speed is slowed to 80 %: power ∝ N^3 → 3 × 0.8^3 = **1.54 kW** (49 % saving) – the reason why VFDs are used.

Short answers (2 marks)

- **Define effective temperature.** Temperature of still saturated air giving the same sensation as the actual air state.
- **What is HEPA?** Filter with ≥ 99.97 % efficiency for 0.3 m particles.
- **Why do forward-curved fans overload?** The power curve keeps rising with flow, so the motor can overload at high Q.
- **Difference between refrigerator and heat pump?** Refrigerator desired effect is Q_L, heat pump desired effect is Q_H; $COP_{HP} = COP_R + 1$.
- **What is ISEER?** Seasonal cooling output / energy input over the Indian cooling season; used for BEE stars.
- **Which humidifier is isothermal?** Steam humidifier.
- **What is the function of eliminator plates?** To prevent water droplets from carrying over into the duct.

Accessories in air-conditioning ducts

- **Dampers** (volume control, fire, smoke dampers), **vibration isolators / flexible joints** for fans, **silencers / acoustic lining** (noise control), **turning vanes** in bends, **grilles / diffusers**, **access doors**, **air curtains** at entrances.

8. Quick Revision and Exam Questions

- Humidifier: spray (adiabatic), steam (isothermal), ultrasonic. Dehumidifier: cooling coil, silica gel / LiCl, desiccant wheel.
- Filters: dry, viscous, wet, ESP, HEPA (99.97 % at 0.3 m), carbon.
- Fan laws: $Q \propto N$, $\Delta P \propto N^2$, $P \propto N^3$; $P = Q\Delta P/\eta$.
- $M - W = E + R + C + S$; ET = still saturated air with the same feeling; comfort ≈ 22-26 °C, 40-60 % RH.
- $COP_{HP} = COP_R + 1$; sources: air, water, ground, solar, waste heat.
- ISEER: higher is better; 5-star split ≥ 5.6 (2026-28).

Likely questions

- Explain different types of humidifiers and dehumidifiers with neat sketches. (long)
- Types of air filters – working and applications; HEPA, ESP. (long)
- Classify fans and blowers; fan laws; fan characteristic curves. (long)
- Define effective temperature; explain the comfort chart; factors affecting comfort. (long)
- Heat pump – working, sources, circuits; compare with refrigerator. (long)
- Energy efficient AC – star rating, ISEER and the features that improve efficiency. (short / long)